

FLNC reads at the boundary of resolvability: what minimap2 and HMM actually do

minimap2 does not 'fail' here – it correctly reports MAPQ=0 with a set of tied alignments. HMM at variant positions does not break those ties either; within a VG equivalence class, members are read-indistinguishable by Theorem 1. Both methods agree on the equivalence class. Bar fill colors indicate VG equivalence-class membership. Read is correctly resolved TO ITS CLASS – assignment within a class collapses by mathematical necessity.

